A Distributed Big Data Analytics Model for Traffic Crash Severity Analysis Using PySpark

Saurav Upadhyaya[1*]

[1]Louisiana State University, LA, United States
*Corresponding Author: supadh10@lsu.edu

Project Description

In this project, I used crash-relevant datasets (~ 2GB) such as crashes, people, vehicles, and Vision Zero traffic fatalities owned by the Chicago Police Department (Data Owner). For processing big data, I used PySpark to process and analyze traffic crashes data. In this application, I used the "apache-spark" image (link) for managing one Master and two Workers. This is done to simplify the dependency management and to have the same runtime environment for all nodes. 1 Spark Master and 2 Spark Workers are the core components in Docker containers.

In this project, I have created a docker-compose.yaml, which includes three services (one spark-master, one spark-worker-1, and one spark-worker-2. In each of these services, I included several properties, such as "build" to reference the Dockerfile and image for referencing the "apache-spark" image. In each service, I have also used several other properties, such as container name, host name, ports, volumes (for mounting the host directory), and a default command that runs once the container is started from the image.

Development Details

- Prerequisites: Docker, Docker desktop, Python, Vim
- Core components. Master VM, Cluster Manager, Worker VMs
- Other Components: Docker Image, Docker Desktop Application, Virtual Machines (each of 6 GB), Spark Driver, Vim, Visual Studio

System Design

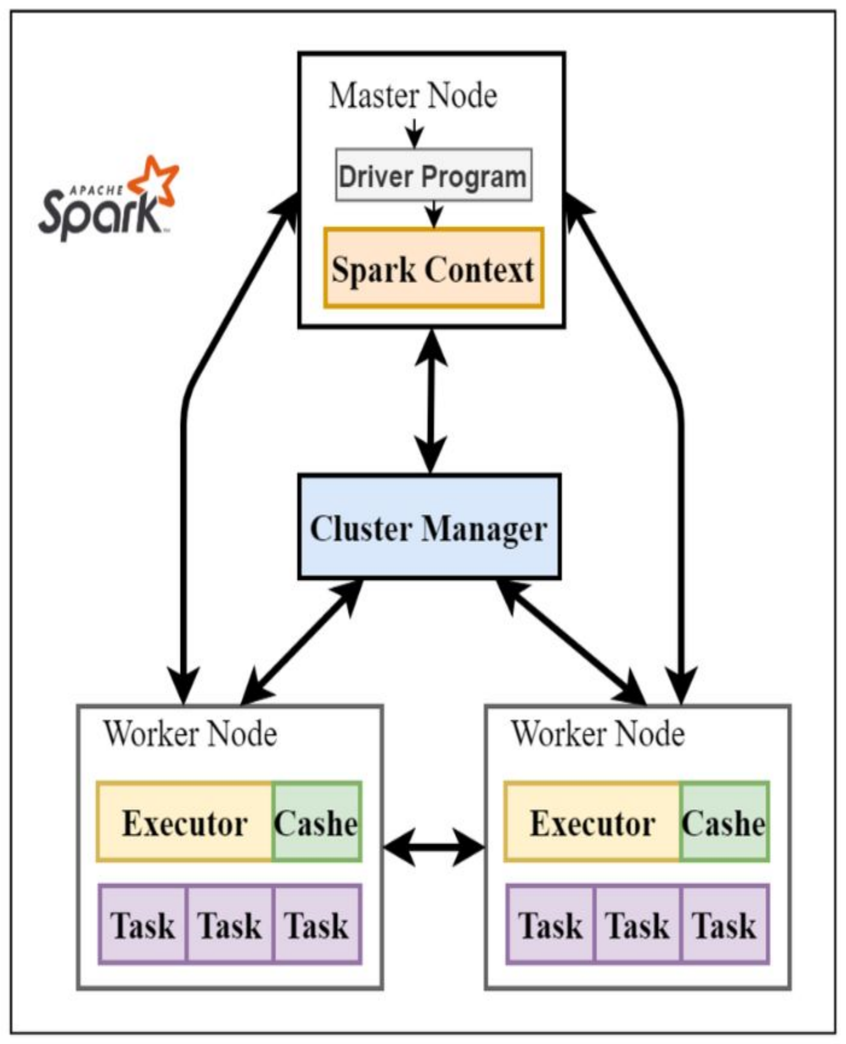


Fig. 1. Overview of Spark Framework

Image Credit: Ullah, Rahmat & Arslan (2020)

Big Data Frameworks

Apache Spark

Main Results & Screenshots

- By creating the "Total number of crashes happening in each city" geospatial visualization (link), I found the number of crashes happening for city in Chicago.
- By creating "Fatal Crashes: Cyclists and Pedestrians" (link) visualization, I found the cities representing the deaths of the cyclists and pedestrians.
- By creating the "Fatal Traffic Accidents Analysis" (link) visualization, I see that the fatal crashes are clustered in certain areas in which age groups and road conditions are key factors.

Major geospatial visualizations obtained from data analysis using PySpark:
Total number of crashes happening in each city (link), Fatal Crashes: Cyclists & Pedestrians (link), Fatal Traffic Accidents Analysis (link)

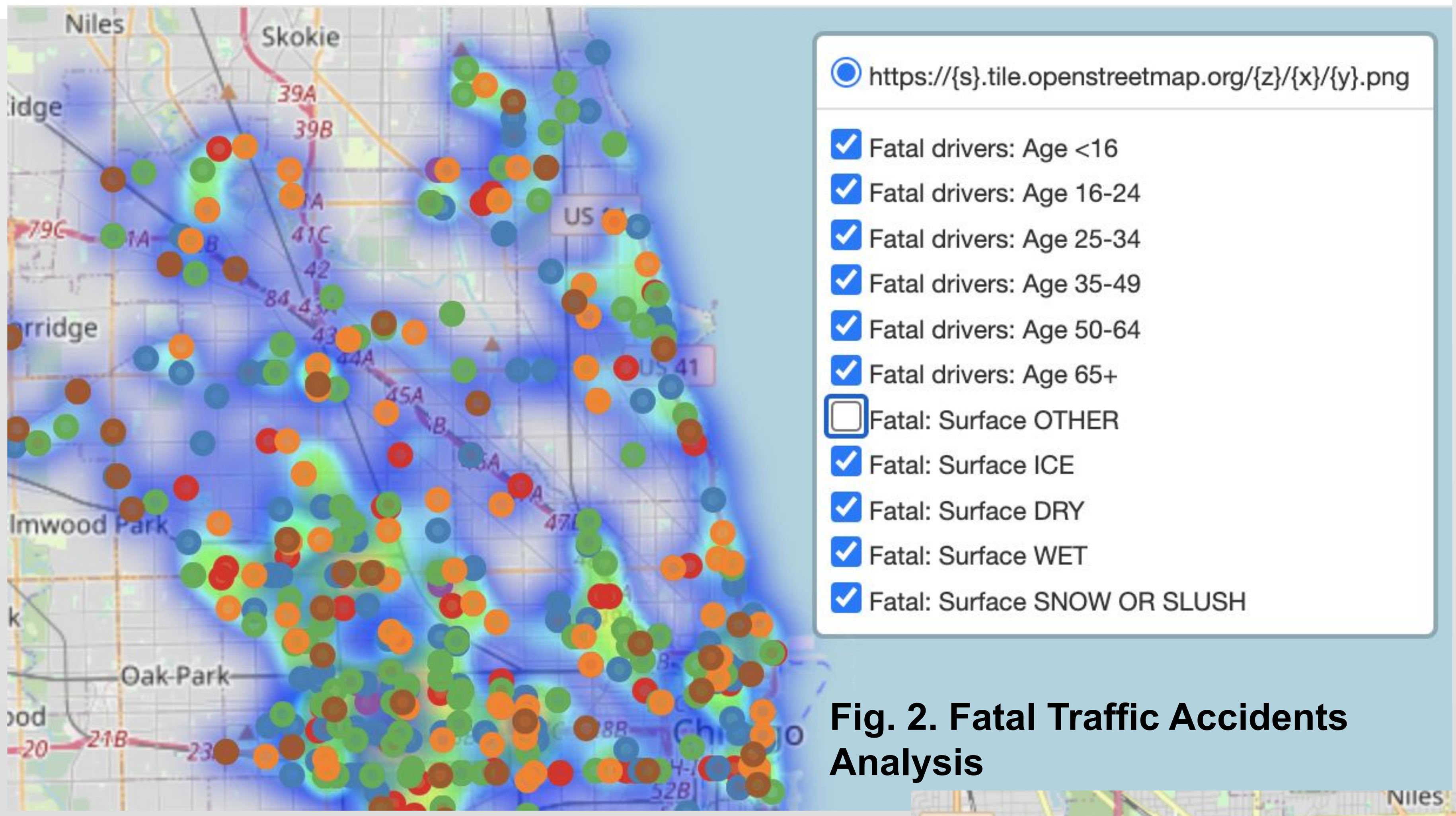


Fig. 2. Fatal Traffic Accidents Analysis

Fig. 3. Fatal crashes: Pedestrians and Cyclists

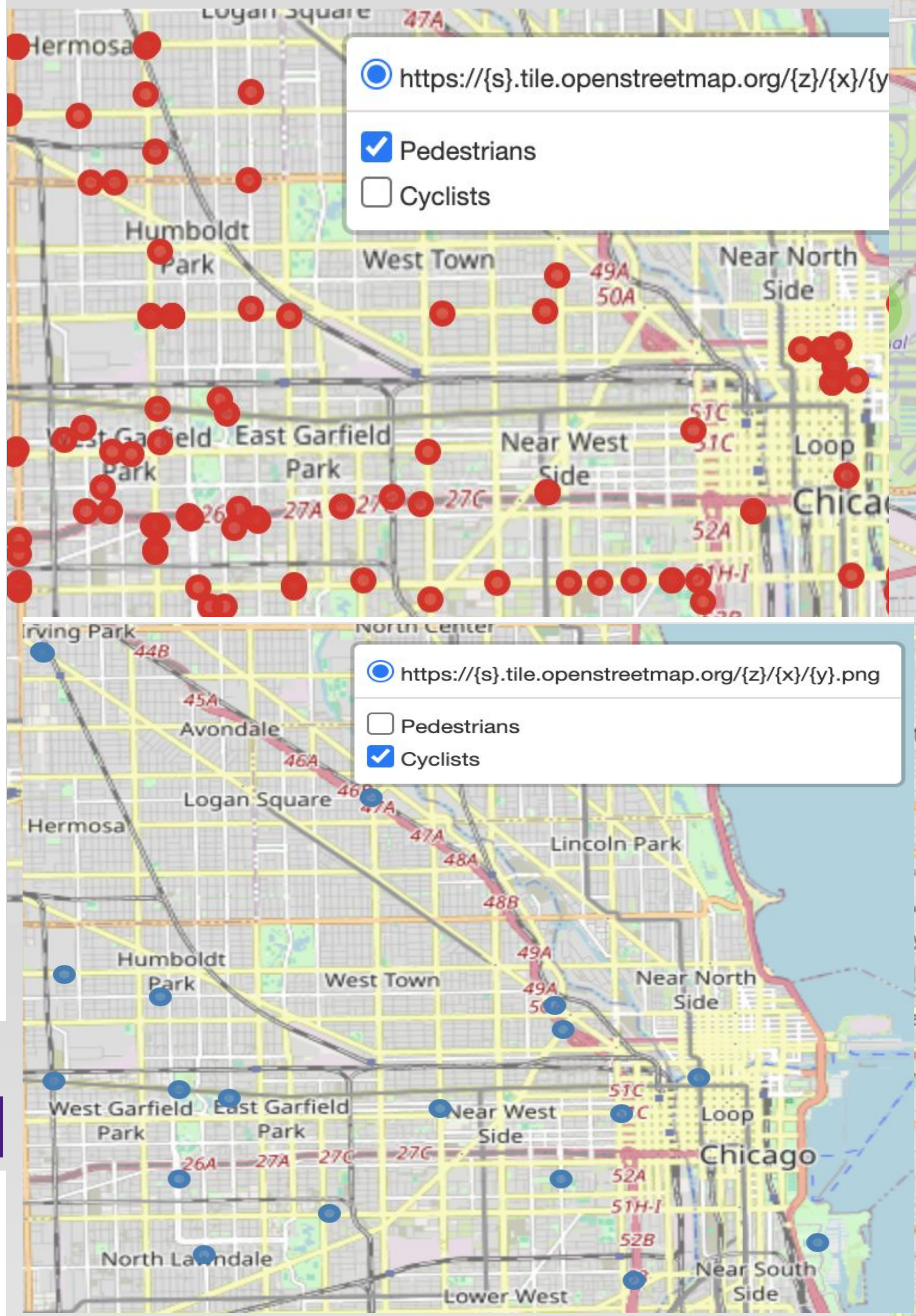
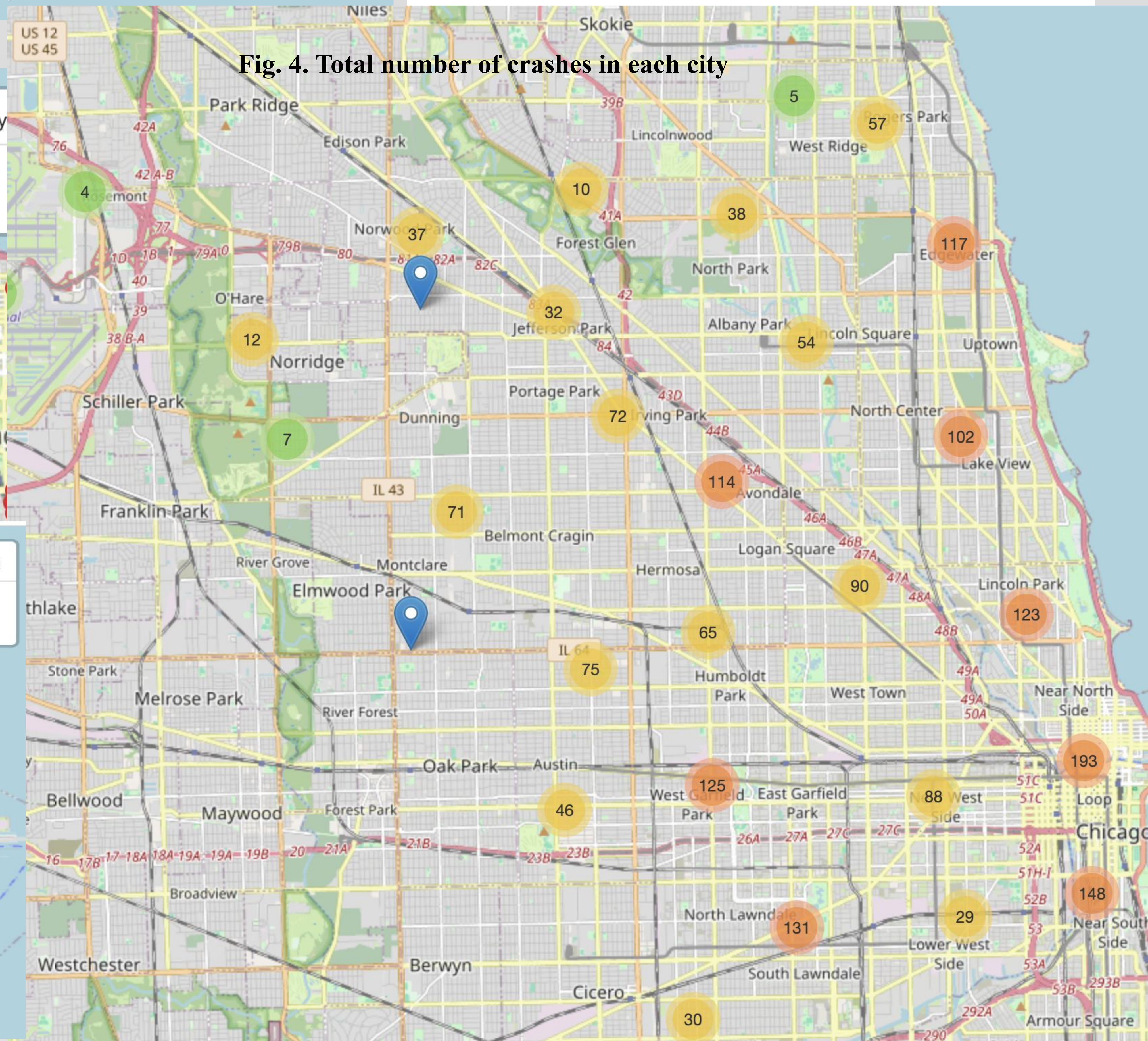


Fig. 4. Total number of crashes in each city



Chicago Traffic Crashes Datasets

crashes (link), vehicles (link), people (link), vision zero fatalities (link)

Conclusion & Future Work

In this project, I used PySpark on Traffic Crashes dataset (2 GB) for distributed data processing and geo-spatial visualizations. I used Apache Spark docker image to create one VM and two Worker VMs (6 GB each). When, Master gets the task, it distributes the task to its workers based on their availability.

On analyzing crashes dataset, I observed that 100k+ crashes are happening every year in Chicago. By creating the crashes markers geospatial visualization, I found the number of crashes happening per city in Chicago. Likewise, by creating fatal geo-spatial visualization, I was able to analyze the geographic locations of the fatal car crashes by examining two important factors, such as age group of the driver and the road surface conditions at the time of the crash. Similarly, by creating fatal heavy hit and run geo-spatial visualization, I found the hotspots where the fatal crashes made by heavy vehicles and hit-run happen frequently.

Thus, these visualizations provide recommendations aiming to improve the road safety. In the future, I am planning to use ML models to forecast the crash prone area and analyze the real-time data of other states as well.